LABORATORY MANUAL

BCSL456D – Technical Writing Using LaTeX

Prepared By

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RAJEEV INSTITUTE OF TECHNOLOGY

HASSAN-573201

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3.	Develop a LaTeX script to create a simple title page of the VTU project Report [Use suitable Logos and text formatting]
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5.	Develop a LaTeX script to create a document that contains the following table with proper labels.

S.No	USN	Student Name	Marks		
			Subject1	Subject2	Subject3
1	4XX22XX001	Name 1	89	60	90
2	4XX22XX002	Name 2	78	45	98
3	4XX22XX003	Name 3	67	55	59

6.	Develop a LaTeX script to include the side-by-side graphics/pictures/figures in the document by using the subgraph concept
7.	Develop a LaTeX script to create a document that consists of the following two mathematical equations

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a} = \frac{-2 \pm \sqrt{2^2-4*(1)*(-8)}}{2*1} = \frac{-2 \pm \sqrt{4+32}}{2}$$

$$\varphi_\sigma^\lambda A_t = \sum_{\pi \in C_t} \operatorname{sgn}(\pi)\varphi_\sigma^\lambda \varphi_\pi^\lambda = \sum_{\tau \in C_{\sigma t}} \operatorname{sgn}(\sigma^{-1}\tau\sigma)\varphi_\sigma^\lambda \varphi_{\sigma^{-1}\tau\sigma}^\lambda = A_{\sigma t}\varphi_\sigma^\lambda$$

8.	Develop a LaTeX script to demonstrate the presentation of Numbered theorems, definitions, corollaries, and lemmas in the document
9.	Develop a LaTeX script to create a document that consists of two paragraphs with a minimum of 10 citations in it and display the reference in the section
10.	Develop a LaTeX script to design a simple tree diagram or hierarchical structure in the document with appropriate labels using the Tikz library
11.	Develop a LaTeX script to present an algorithm in the document using algorithm/algorithmic/algorithm2e library
12.	Develop a LaTeX script to create a simple report and article by using suitable commands and formats of user choice.

1. **Develop a LaTeX script to create a simple document that consists of 2 sections [Section1, Section2], and a paragraph with dummy text in each section. And also include header [title of document] and footer [institute name, page number] in the document.**

CODE:

```
\documentclass[12pt,a4paper]{article}
\usepackage[left=2cm,right=2cm,top=2cm,bottom=2cm]{geometry}
\usepackage{fancyhdr}
\begin{document}

\title{ Different type of cuisines}

\fancyhf{}
\fancyhead[R]{Different type of cuisines}
\fancyfoot{}
\fancyfoot[L]{Rajeev Institute of Technology, Hassan}
\renewcommand{\footrulewidth}{0.4pt}
\fancyfoot[R]{\thepage}
\fancypagestyle{plain}{
        \fancyhead{}
        \fancyhead[R]{Different type of cuisines}
        \fancyfoot{}
        \fancyfoot[L]{Rajeev Institute of Technology, Hassan}
    \renewcommand{\footrulewidth}{0.4pt}
    \fancyfoot[R]{\thepage}
        }
\pagestyle{fancy}
\maketitle

\section{Indian cuisine}
India's unique blend of cuisines evolved through large-scale cultural interactions with neighboring Persia, ancient Greece, Mongols and West Asia. New World foods such as chili peppers, tomatoes, potatoes and squash, introduced by Arab and Portuguese traders during the sixteenth century, and European cooking styles introduced during the colonial period added to the diversity of Indian cuisine.

\section{ Mexican cuisine }
The Mexican culture is diverse and vibrant, where they are proud of their native heritage. They celebrate their festivals with music and dance and they have their folk dance too! The way they celebrate regional festivals with their own customs is very similar to India. The vibrancy and diversity of the Mexican culture reflect in their food. They use fresh produce in their food like chile peppers, herbs and vegetables, cheese, tomatoes and beans. Mexico shares a distinctly similar heritage and food culture with India and therefore, Indians too relish the Mexican food.\\

Mexican cuisine is ever-evolving as its staples continue to change. First, the staple ingredients of Mexican cuisine were corns and bean and slowly, the new ingredients in the form of spices were added over the past 600 years to make Mexican cuisine what it is today.  \\
```

When we think of Mexican cuisine, we immediately think of cheesy nachos and enormous fishbowl margaritas. But Mexican cuisine is much more than that. The famous tortilla, taco, flan, enchiladas, and more are very much a part of Mexican cuisine.

```
\section{Thai Cuisine}
A blend of eastern and western influences, Thai food harmonizes the dietary preferences of many cultures into a synthesis that blends sweet, sour, spicy, and bitter flavors to create a unique dining experience. Thai cuisine boasts some of the most popular dishes in the world, but as delicious as they are the story of their origins is just as fascinating. Today let us delve into this history and see how some of your favorite Thai recipes came to be. \\

\section{ Italian Cuisine }
Like any other good cooking, Italian cooking is all about ingredients. Many Italian dishes feature a few key ingredients. The first course may consist of pasta or rice with a seasonal ingredient like tomato or another vegetable. A typical second course may consist of meat or fish prepared with seasonal ingredients. Therefore, the ingredients you use will determine the quality of the Italian dish you prepare for your friends or family. However, there are various tips for choosing the right ingredients for your Italian dish.

\begin{itemize}
\item Diverse nutrient profiles: Different cuisines offer a wide variety of nutrients, ensuring a balanced diet when consumed in moderation.
\item Heart-healthy options: Many cuisines, such as Mediterranean and Japanese, include ingredients that promote heart health, such as olive oil, fish, and whole grains.
\item Focus on fresh ingredients: Several cuisines prioritize fresh, seasonal ingredients, ensuring optimal flavor and nutritional value.
\item Promotion of portion control: Some cuisines, like Japanese, encourage portion control through the presentation of smaller servings of a variety of dishes.

Exploring different cuisines allows for a rich cultural experience and promotes social interaction through shared meals with family and friends. This aspect enhances enjoyment and appreciation of diverse culinary traditions.

\end{itemize}


\end{document}
```

2. **Develop a LaTeX script to create a document that displays the sample Abstract/Summary**

CODE:

```
\documentclass[10pt,a4paper]{article}
\usepackage[utf8]{inputenc}
\usepackage{amsmath}
\usepackage{amsfonts}
\usepackage{amssymb}
\usepackage[left=3cm,right=3cm,top=2cm,bottom=2cm]{geometry}
\begin{document}
\thispagestyle{plain}

\begin{center}
\Large
\textbf{COVID-19 and Comorbid}\\
\vspace{0.4cm}
\large AI in Medical Field\\
\vspace{0.9cm}
\textbf{Abstract}
\end{center}
The advent of COVID-19 marks a significant turning point in medical history, leading to
increased reliance on advanced technologies such as deep learning models. These models
have become indispensable for promptly identifying COVID-19 in medical images
obtained through Computerized Tomography (CT), Ultrasound, and X-ray scans. This
study aims to equip healthcare professionals with an additional tool to improve the
formulation of effective treatment plans and containment strategies for the disease. A new
hybrid architecture called MobNetCov19 has been introduced for diagnosing COVID-19
in patients with comorbidities, utilizing a Convolutional Neural Network (CNN) model.
The research investigates optimized versions of VGG19, ResNetRS152, and
MobNetCov19 (VRM) models, facilitating a comparative assessment of different imaging
modalities. Despite the limited availability of COVID-19 datasets, sophisticated detection
models have been carefully developed.
\\
Keywords: Convolutional Neural Network, Computerized Tomography, Coronavirus-19,
MobNetCov19, VRM.
\\
\textbf{Summary}

The study reveals that CT scans excel in disease detection compared to X-ray and
Ultrasound scans. Researchers tested three models—VGG19, ResNetRS152, and
MobNetCov19—Carefully adjusting parameters for accuracy. Augmentation techniques
significantly improved the models' precision, particularly in early disease detection.
Limited data availability posed a challenge for training more complex models. Results
indicate that CT and Ultrasound datasets outperform X-ray data in disease detection. The
MobNetCov19 model, fine-tuned rigorously, achieved impressive accuracy rates of 98\%
for CT, 98\% for Ultrasound, and 96\% for X-Ray data.
\end{document}
```

3. **Develop a LaTeX script to create a simple title page of the VTU project Report [Use suitable Logos and text formatting]**

CODE:

```
\documentclass[12pt, a4paper]{article}
\usepackage[utf8]{inputenc}
\usepackage{times}
\usepackage{tabularx}
\usepackage{graphicx}
\usepackage[top=1in,bottom=1in,left=1.5in,right=1.25in]{geometry}

\begin{document}
\pagestyle{empty}
\vspace{1.5 cm}
\begin{center}
\centering \textbf{\Large Visvesvaraya Technological University }
\end{center}
\centering {\footnotesize “Jnana Sangama”, Belagavi, Karnataka-590018}
\vspace{0.5cm}
\begin{figure}[ht]
  \centering
  {\includegraphics[width=4cm]{V TU LOGO.jpg}}

\end{figure}

\vspace{0.5cm}

\large{\textbf{“A Random Walk into \LaTeX ”}}\\ \vspace{0.2cm}
\begin{center}

\small A project report submitted to \\Visvesvaraya Technological University. Belgaum, Karnataka \\
\textit{in the partial fulfillment of the requirements for the award of degree of} \\
\end{center}
\begin{center}
% \vspace{0.25 cm}

\begin{normalsize}
\textbf{\textit{Bachelor of Engineering }} \\
in \\
\textbf{\textit{Computer Science and Engineering}} \\
by \\
\end{normalsize}


\end{center}



\vspace{0.1cm}
\begin{center}
```

```
\begin{tabular}{ c c c c c c }
\noindent

\textnormal{Name} & & &  \textnormal{USN}\\

\end{tabular}
\end{center}
\vspace{0.5cm}
\small Under the guidance of\\ {\bf Ms. Nisha A G  }\\
Assistant Professor\\
\begin{figure}[ht]
   \centering
   {\includegraphics[width=4cm]{college_logo.jpg}}
\end{figure}

\small Department of Computer Science and Engineering\\ \small Rajeev Institute of
Technology\\ \small Hassan

\small 2023-2024\\
\vspace{0.5cm}

\end{document}
```

4. **Develop a LaTeX script to create the Certificate Page of the Report [Use suitable commands to leave the blank spaces for user entry]**

```
\documentclass[12pt, a4paper]{article}
\usepackage[utf8]{inputenc}
\usepackage{times}
\usepackage{tabularx}
\usepackage{graphicx}
\usepackage[top=1in,bottom=1in,left=1.5in,right=1.25in]{geometry}

\begin{document}
\pagestyle{empty}

\begin{center}
\vspace{0.1 cm}
\Large\textbf{RAJEEV INSTITUTE OF TECHNOLOGY}\\
\small HASSAN 573217-KARNATAKA \\

\end{center}
\begin{figure}[ht]
   \centering
   \includegraphics[width=6cm]{college_logo.jpg}
\end{figure}
\begin{center}
   \bf DEPARTMENT OF COMPUTER SCIENCE \& ENGINEERING
\end{center}
\begin{center}

{\large{\bf{CERTIFICATE}}}
\end{center}
%\begin{center}
This is to certify that the work entitled
%\end{center}
\small{
\textbf{“Hospital Management System”} carried out by \textbf{Mr./Ms.XYZ },\bf{ USN} a bonafide student of \textbf{RAJEEV INSTITUTE OF TECHNOLOGY, HASSAN }in partial fulfilment for the award of \textbf{BACHELOR OF ENGINEERING} in \textbf{COMPUTER SCIENCE AND ENGINEERING} of the\textbf{ VISVESVARAYA TECHNOLOGICAL UNIVERSITY, BELAGAVI} during the year \textbf{2023 - 2024}. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the Report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said Degree.}

\vspace{1.3cm}
\begin{center}


\begin{tabular}{c c c c c c c c c c c c } \\
  \textbf{Ms. Nisha A G}  & & & & &  \textbf{Dr. Myna A N} & & & & & \textbf{Dr. M Venu Gopala Rao}\\
```

```
  Project Guide  & & & & & Professor \& Head & & & & & Principal \\
  Dept. of CSE  & & & & & Dept. of CSE  & & & & & NCE Hassan
\end{tabular}
\end{center}
\begin{flushleft}



\vspace{1cm}
\begin{tabular}{ c c c c c c }
  Name of the Examiners & & & & & Signature with Date  \\
 & & & & &  \\
1.............................................& & & & & \\
  & & & & &  \\
2. ......................................& & & & &  \\
 & & & & &  \\

\end{tabular}
\end{flushleft}
\end{document}
```

5. **Develop a LaTeX script to create a document that contains the following table with proper labels.**

S.No	USN	Student Name	Marks		
			Subject1	Subject2	Subject3
1	4XX22XX001	Name 1	89	60	90
2	4XX22XX002	Name 2	78	45	98
3	4XX22XX003	Name 3	67	55	59

CODE :

```
\documentclass[10pt,a4paper]{article}
\usepackage[utf8]{inputenc}
\usepackage{amsmath}
\usepackage{amsfonts}
\usepackage{amssymb}
\usepackage[left=2cm,right=2cm,top=2cm,bottom=2cm]{geometry}
\usepackage{multirow}
\begin{document}

\begin{center}
\begin{Large}
\textbf{Table: Students Details of 4th semester CSE}
\end{Large}
\end{center}

\section*{Marks Details}

\begin{tabular}{|c|c|c|c|c|c|}
\hline
\multirow{2}{*}{S.No} & \multirow{2}{*}{USN} & \multirow{2}{*}{Student
Name} & \multicolumn{3}{c|}{Marks} \\
\cline{4-6}
 & & & Subject1 & Subject2 & Subject3 \\
\hline
1 & 4XX22XX001 & Name 1 & 88 & 77 & 97 \\
\hline
2 & 4XX22XX002 & Name 2 & 74 & 78 & 66 \\
\hline
3 & 4XX22XX003 & Name 3 & 88 & 82 & 79 \\
\hline
\end{tabular}

\end{document}
```

6. **Develop a LaTeX script to include the side-by-side graphics/pictures/figures in the document by using the subgraph concept.**

CODE:

```
\documentclass{article}
\usepackage{graphicx}
\usepackage{subcaption}% For side-by-side
\begin{document}
\textbf{\huge Side-by-Side Subgraph}
\begin{figure}[ht]
\subfloat[Signal Graph ]{
\begin{minipage}[c][1\width]{0.3\textwidth}
\centering
\includegraphics[width=1\textwidth]{signal graph.png}
\end{minipage}}
\hfill
\subfloat[Line Graph]{
\begin{minipage}[c][1\width]{
0.3\textwidth}
\centering
\includegraphics[width=1.1\textwidth]{GRAPHS.png}
\end{minipage}}
\hfill
\subfloat[Bar Graph]{
\begin{minipage}[c][1\width]{
0.3\textwidth}
\centering
\includegraphics[width=1.2\textwidth]{bar graph.png}
\end{minipage}}
\caption{Different types of graphs}
\end{figure}
\end{document}
```

7. **Develop a LaTeX script to create a document that consists of the following two mathematical equations**

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-2 \pm \sqrt{2^2 - 4 \cdot 1 \cdot (-8)}}{2 \cdot 1} \\ &= \frac{-2 \pm \sqrt{4 + 32}}{2} \end{aligned} \quad \quad \quad \& \quad \begin{aligned} \varphi_{\sigma}^{\lambda} A_{t} &= \sum_{\pi \in C_{t}} \operatorname{sgn}(\pi) \varphi_{\sigma}^{\lambda} \varphi_{\pi}^{\lambda} \\ &= \sum_{\tau \in C_{\sigma t}} \operatorname{sgn}(\pi^{-1} \tau \sigma) \varphi_{\sigma}^{\lambda} \varphi_{\pi^{-1} \tau \sigma}^{\lambda} \\ &= A_{\sigma t} \varphi_{\sigma}^{\lambda} \end{aligned}$$

CODE:

```
\documentclass{article}
\usepackage{amsmath} % This package is required for the 'aligned' environment
\begin{document}
\title{Following Two Mathematical Equations}
\maketitle
% Use the 'align' environment to align equations properly
\[
\begin{aligned}
% Left side: equations about 'x'
\begin{split}
x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\
&= \frac{-2 \pm \sqrt{2^2 - 4 \cdot 1 \cdot (-8)}}{2 \cdot 1} \\
&= \frac{-2 \pm \sqrt{4 + 32}}{2}
\end{split}
\quad \quad \quad
&
% Right side: equations about 'phi_{\sigma}^{\lambda} A_t'
\begin{split}
\varphi_{\sigma}^{\lambda} A_{t} &= \sum_{\pi \in C_{t}}
\operatorname{sgn}(\pi)
\varphi_{\sigma}^{\lambda} \varphi_{\pi}^{\lambda} \notag \\
&= \sum_{\tau \in C_{\sigma t}} \operatorname{sgn}(\pi^{-1} \tau \sigma)
\varphi_{\sigma}^{\lambda} \varphi_{\pi^{-1} \tau \sigma}^{\lambda} \notag \\
&= A_{\sigma t} \varphi_{\sigma}^{\lambda}
\end{split}
\end{aligned}\]
\end{document}
```

8. **Develop a LaTeX script to demonstrate the presentation of Numbered theorems, definitions, corollaries, and lemmas in the document**

CODE:

```
\documentclass{article}
\usepackage{amsthm}
% Define theorem-like environments
\newtheorem{theorem}{Theorem}
\newtheorem{definition}{Definition}
\newtheorem{corollary}{Corollary}
\newtheorem{lemma}{Lemma}
\begin{document}
\title{Numbered Theorems, Definitions, Corollaries, and Lemmas in the Document}
\maketitle
\begin{theorem}
(Pythagorean Theorem) In a right-angled triangle, the square of the length of the hypotenuse
is equal to the sum of the squares of the lengths of the other two sides.
\begin{equation}
a^2 + b^2 = c^2
\end{equation}
\end{theorem}
% Presenting a definition with an example
\begin{definition}
(Prime Number) A prime number is a natural number greater than 1 that is not divisible by
any number other than 1 and itself.
\begin{itemize}
\item Example: 2, 3, 5, and 7 are prime numbers.
\end{itemize}
\end{definition}
% Presenting a corollary with an example
\begin{corollary}
(Euclid's Corollary) There are infinitely many prime numbers.
\begin{itemize}
\item Proof: Assume there are finitely many primes. Let them be $p_1, p_2, \ldots,p_n$.
Consider the number $N = p_1 \cdot p_2 \cdots p_n + 1$.
This number is not divisible by any of the primes $p_1$ through $p_n$.
Therefore, there must be a prime factor not in the list, contradicting the assumption that there
are only finitely many primes.
\end{itemize}
\end{corollary}
% Presenting a lemma with an example
\begin{lemma}
(Basic Arithmetic Identity) For any real numbers $a$ and $b$, we have: 17
\begin{equation}
(a + b)^2 = a^2 + 2ab + b^2.
\end{equation}
\end{lemma}
\end{document}
```

9. Develop a LaTeX script to create a document that consists of two paragraphs with a minimum of 10 citations in it and display the reference in the section

CODE:

```
\documentclass{article}
\usepackage[numbers]{natbib} % Use the 'natbib' package for citation management

\begin{document}
\title{Document that Consists of Two Paragraphs with a minimum of 10 Citations in it and Displaying the References in the Section}
\maketitle

% Write two paragraphs with at least 10 citations
\paragraph{Paragraph 1}
The theory of relativity has been explored extensively in various scientific papers \citep{author1, author2, author3}. Einstein's contributions to physics are profound and have paved the way for many modern discoveries \citep{author4, author5}. Researchers continue to investigate the complexities of spacetime and the universe \citep{author6, author7}. These advancements have led to new methods of measurement and analysis in cosmology \citep{author8}.

\paragraph{Paragraph 2}
Recent studies have focused on the impact of climate change on various ecosystems \citep{author9, author10}. Scientists are examining how rising temperatures and changing weather patterns affect biodiversity and human health \citep{author11, author12}. Furthermore, interdisciplinary research has brought about innovative solutions for sustainable development \citep{author13}. Collaboration among experts from different fields is essential to address the challenges of global warming \citep{author14}.

% Add a references section
\newpage % Optional: Start the references on a new page
\begin{thebibliography}{99}
\bibitem{author1} Author One. \textit{Title of Article One}. Journal Name, vol. 10, no. 1, pp. 1--10, 2022.
\bibitem{author2} Author Two. \textit{Title of Book Two}. Publisher Name, 2021.
\bibitem{author3} Author Three. \textit{Title of Conference Paper Three}. In: Proceedings of Conference, pp. 100--110, 2020.
\bibitem{author4} Author Four. \textit{Title of Article Four}. Journal Name, vol. 9, no. 2, pp. 15--25, 2019.
\bibitem{author5} Author Five. \textit{Title of Book Five}. Publisher Name, 2018.
\bibitem{author6} Author Six. \textit{Title of Conference Paper Six}. In: Proceedings of Conference, pp. 200--210, 2017.
\bibitem{author7} Author Seven. \textit{Title of Article Seven}. Journal Name, vol. 8, no. 3, pp. 30--40, 2016.
\bibitem{author8} Author Eight. \textit{Title of Book Eight}. Publisher Name, 2015.
\bibitem{author9} Author Nine. \textit{Title of Conference Paper Nine}. In: Proceedings of Conference, pp. 300--310, 2014.
\bibitem{author10} Author Ten. \textit{Title of Article Ten}. Journal Name, vol. 7, no. 4, pp. 50--60, 2013.
\bibitem{author11} Author Eleven. \textit{Title of Article Eleven}. Journal Name, vol. 6, no.
```

```
5, pp. 70--80, 2012.
\bibitem{author12} Author Twelve. \textit{Title of Article Twelve}. Journal Name, vol. 5,
no. 6, pp. 90--100, 2011.
\bibitem{author13} Author Thirteen. \textit{Title of Book Thirteen}. Publisher Name, 2010.
\bibitem{author14} Author Fourteen. \textit{Title of Conference Paper Fourteen}. In:
Proceedings of Conference, pp. 400--410, 2009.
\end{thebibliography}
\end{document}
```

10. Develop a LaTeX script to design a simple tree diagram or hierarchical structure in the document with appropriate labels using the Tikz library

CODE:

```
\documentclass{article}
% Import the 'tikz' package for creating diagrams
\usepackage{tikz}
\usetikzlibrary{trees}
\begin{document}
\title{Simple Tree Diagram or Hierarchical Structure in the Document with
appropriate
Labels using the Tikz Library}
\maketitle
% Create a tree diagram
% Wrap the TikZ picture in a 'center' environment to center the diagram

\begin{center}
\begin{tikzpicture} [
level 1/.style={sibling distance=7cm},
level 2/.style={sibling distance=4cm},
level distance=2cm,
every node/.style={font=\large} % Increase font size here (e.g. \large, \Large)
]
% Root node
\node [red] {Root}
% Level 1 nodes
child { node [cyan] {Child 1}  [cyan]
% Level 2 nodes
child { node [cyan] {Grandchild 1.1} }
child { node [cyan] {Grandchild 1.2} }
}
child { node [magenta] {Child 2} [magenta]
% Level 2 nodes
child { node [magenta] {Grandchild 2.1} }
child { node [magenta] {Grandchild 2.2} }
};
\end{tikzpicture}
\end{center}
\end{document}
```

11. Develop a LaTeX script to present an algorithm in the document using algorithm/algorithmic/algorithm2e library.

CODE:

```
\documentclass{article}
\usepackage[ruled, linesnumbered]{algorithm2e}

\begin{document}
\begin{algorithm}[H]
\SetAlgoLined
\SetKwFunction{Dijkstra}{Dijkstra}
\SetKwProg{Fn}{Function}{:}{}
\Fn{\Dijkstra{$G, s$}}{
$d[s] \gets 0$\;
\ForEach{$v \in V$}{
$d[v] \gets \infty$\;
$prev[v] \gets$ undefined\;
}
$Q \gets V$\;
\While{$Q$ is not empty}{
$u \gets$ vertex in $Q$ with minimum $d[u]$\;
Remove $u$ from $Q$\;
\ForEach{$v \in$ neighbors of $u$}{
$alt \gets d[u] + \mathrm{weight}(u, v)$\;
\If{$alt < d[v]$}{
$d[v] \gets alt$\;
$prev[v] \gets u$\;
}
}
}
\KwRet{$d[], prev[]$}\;
}
\caption{Dijkstra's Algorithm}
\label{algo:dijkstra}
\end{algorithm}
\end{document}
```

12. Develop a LaTeX script to create a simple report and article by using suitable commands and formats of user choice.

CODE: <u>ARTICLE</u>

```
\documentclass[conference]{IEEEtran}
\IEEEoverridecommandlockouts
% The preceding line is only needed to identify funding in the first footnote. If that is unneeded,
please comment it out.
\usepackage{cite}
\usepackage{amsmath,amssymb,amsfonts}
\usepackage{algorithmic}
\usepackage{graphicx}
\usepackage{float}
\usepackage{array, caption, tabularx,  ragged2e,  booktabs}
\usepackage{tabu}
\usepackage{textcomp}
\usepackage{xcolor}
\def\BibTeX{{\rm B\kern-.05em{\sc i\kern-.025em b}\kern-.08em
   T\kern-.1667em\lower.7ex\hbox{E}\kern-.125emX}}

\usepackage[british]{babel}
\usepackage{csquotes}
\usepackage[style=apa,sorting=none,backend=biber,citestyle=numeric]{biblatex}
\DeclareLanguageMapping{british}{british-apa}

\DeclareFieldFormat{labelnumberwidth}{\mkbibbrackets{#1}}

\defbibenvironment{bibliography}
  {\list
    {\printtext[labelnumberwidth]{%
     \printfield{labelprefix}%
     \printfield{labelnumber}}}
    {\setlength{\labelwidth}{\labelnumberwidth}%
     \setlength{\leftmargin}{\labelwidth}%
     \setlength{\labelsep}{\biblabelsep}%
     \addtolength{\leftmargin}{\labelsep}%
     \setlength{\itemsep}{\bibitemsep}%
     \setlength{\parsep}{\bibparsep}}%
     \renewcommand*{\makelabel}[1]{\hss##1}}
  {\endlist}
  {\item}

% \usepackage[natbib=true,backend=biber,sorting=nyt,style=apa]{biblatex}
% \renewcommand*{\bibfont}{\fontsize{12}{14}\selectfont}

% add your references to this file
\addbibresource{Ref.bib}
```

```
\begin{document}

\title{LQR based Optimal Controller for Antilock Braking System \\}

\author{
\IEEEauthorblockN{ Name}
\IEEEauthorblockA{B.E Student \\ Department of CSE\\
\textit{College name} \\
your mail id}
\and
\IEEEauthorblockN{Dinesh S}
\IEEEauthorblockA{\textit{Assistant Professor} \\ Department of CSE  \\
\textit{College name} \\
dineshs@rithassan.ac.in} }
\maketitle

\begin{abstract}
Numerous control techniques have been devised for Anti-Lock Braking Systems (ABS), each
differing in their theoretical foundations and performance in response to varying road conditions.
One of the primary causes of vehicle imbalance and accidents during sudden braking is wheel
lock-up, which compromises maneuverability. The development of ABS aims to prevent such
wheel lock-up, resulting in a significant reduction in fatal accidents. ABS, an abbreviation for
the anti-lock braking system, enhances vehicle control, particularly in instances of abrupt braking
on slippery road surfaces. The objective of this control is to augment wheel traction in the desired
direction while simultaneously ensuring vehicle stability and strength, all while minimizing the
vehicle's stopping distance. In this project, an ABS system employing Optimal Control
Algorithm is implemented. The overarching goal of the anti-lock braking system is to effectively
regulate wheel slip to attain optimal wheel traction and maximize vehicle
deceleration.
\end{abstract}

\section{Introduction}
The Anti-Lock Braking System (ABS) has emerged as an indispensable cornerstone within the
landscape of contemporary automotive safety technology, a sentinel of stability and control when
the brakes are called upon in the most exigent of circumstances. Its paramount importance lies
in its unwavering commitment to averting wheel lock-up during the throes of hard or emergency
braking, a feat that not only safeguards the driver but also those sharing the road. At its core,
ABS embodies a holistic approach to vehicular safety, intertwining a web of technologies and
control strategies, each meticulously orchestrated to combat the perilous loss of steering control
or skidding. In essence, ABS is the embodiment of proactive intervention, a guardian of road
safety that ensures vehicles maintain optimal traction with the road surface, thereby preserving
stability and steering responsiveness.
\newline
ABS maintains the connection between wheels and the steering wheel throughout the braking
process. It achieves this by issuing rapid commands to the wheels, alternating between braking
and releasing brake calipers at extremely short intervals, sometimes up to twenty times per
second. This controlled approach prevents abrupt stops and allows passengers to stay safely
within the vehicle during sudden stops, avoiding potential ejections. ABS also enables tractive
contact between the wheels and the road, as drivers can brake and steer simultaneously without
skidding.
```

\newline
The Anti-Lock Braking System transcends its role as a mere automotive innovation; it stands as a guardian of road safety, a sentinel of stability and control. Its multifaceted attributes, encompassing safety enhancement, efficient braking, and consistent brake force distribution, make it an indispensable feature of modern vehicles. The intricate web of components, control strategies, and mathematical modelling, combined with its adaptive nature, underscores its role as a linchpin in vehicular safety. ABS continues to evolve, poised at the nexus of advanced sensor technologies, seamlessly integrating with other safety systems, and exploring innovative control strategies, thus reaffirming its enduring relevance in making our roads safer for all who traverse them.

\section{Literature Survey}
\begin{itemize}
\item Dankan Gowda V et al. proposed a study that discusses the effects of a sliding mode controller-based Anti-lock Braking System (ABS), including the simulation of longitudinal wheel dynamics. Within the context of ABS, this sliding surface typically represents the desired wheel slip ratio, defined as the ratio of the wheel's actual speed to the vehicle's speed. The primary objective of their research is to prevent wheel lockup and maintain vehicle stability during braking. To achieve the desired wheel slip within a specified range, they employ an ABS system that utilizes a bang-bang control law. The optimized value of this study is reported as 0.12. In conclusion, the paper asserts that the SMC controller demonstrates superior performance and efficiency in terms of brake pressure modulation.

\item Andrei Aksjonov and at.all have introduced a research paper that examines an open-loop fuzzy logic-based ABS control approach. This method has been validated to effectively address road-induced disturbances. Leveraging the intricate features of fuzzy control, the system achieves its maximum safety performance. Moreover, this control approach demonstrates a higher degree of feasibility for optimal slip control.

\item Ahmad O. Moaaz and et.all have developed an ABS model utilizing MATLAB simulink and explored different control strategies for their model. Their findings indicated that the utilization of Bang-Bang, PD, and PID controllers resulted in effective braking performance. These controllers concurrently regulate both the vehicle velocity and wheel angular speed, thus preventing skidding during braking maneuvers. Additionally, the PID controller was particularly noteworthy for reducing stopping time and distance, achieving a stopping distance of 23.61 meters in just 2.319 seconds, as opposed to the scenarios without control, with Bang-Bang control, and with PD control. .

\item Gengxin Qi and et.all have introduced an innovative approach for enhancing ABS control by incorporating road recognition technology. Their novel ABS control method involves estimating the road surface adhesion coefficient using a fuzzy logic control technique and subsequently calculating the optimal slip rate based on this estimation. They have developed a robust ABS controller that takes these slip rates into account. The results of their research demonstrate that this controller yields effective control outcomes and has the capability to provide real-time estimations of road adhesion coefficients.

\item Kartik Bhasin and et.all have discussed the significant improvements in vehicle safety brought about by advancements in technology, with a particular focus on the Anti-lock Braking System (ABS). This technology has played a crucial role in reducing the incidence of injuries associated with vehicle automation.However, it's worth noting that ABS alone may

not always effectively address all road conditions. To overcome this limitation, the automotive industry has introduced additional technologies such as Electronic Brakeforce Distribution (EBFD), Electronic Stability Control (ECS), and Traction Control System (TCS). In this paper they delve into the principles and functioning of ABS technology, as well as explore its advanced counterparts and their contributions to enhancing vehicle safety.

\item S. Suresh and et.all have introduced a novel concept called the 'Intelligent Braking System' (IBS) in their work, which relies on a multitude of sensors to promptly respond to emergency situations. This advanced system incorporates an ultrasonic wave emitter positioned at the front of the vehicle, accompanied by an ultrasonic receiver to capture the reflected signal. By analyzing the reflected wave, the system calculates the distance between the vehicle and any obstacles in its path. Subsequently, a microcontroller processes this data and triggers the application of brakes when necessary. The IBS system represents a visionary development in automotive safety, offering a glimpse into the future. With the implementation of the IBS system, we have the potential to prevent a greater number of accidents and, in turn, save more lives on the road.

\item Mansaf Ali Abro and et.all have introduced an innovative approach for designing a novel Anti-lock Braking System (ABS). Their work underscores the importance of incorporating ABS in vehicles to prevent wheel locking and skidding. To achieve this, they developed a system model and derived mathematical equations to describe its behavior. These equations were then implemented using simulink for further analysis. Their research involved running the system model and simulating results in three distinct scenarios, each corresponding to different road surfaces: concrete/dry asphalt, snowy, and icy surfaces. Notably, this study incorporates the use of the standard Pacejka magic formula, adding depth and comprehensiveness to their ABS system

design.
\end{itemize}
\section{METHEDOLOGY AND IMPLEMENTATION}

\subsection{Block Diagram of Proposed System}
\begin{figure}[H]
  \centering
  \includegraphics[height=15cm,width=08cm,keepaspectratio]{Picture7.png}
  \caption{Block Diagram of proposed system }
  \label{fig:Block Diagram}
\end{figure}
Figure 2 shows the block diagram of basic design flow of An Antilock-Braking System The system comprises several key components, including a vehicle speed sensor, wheel speed sensor, Tire-Road interaction module, Machine Learning Control Algorithm, Brake Actuator Valve, and Brake Actuator. The Vehicle vehicle speed sensor is responsible for monitoring the vehicle's speed, while the Wheel speed sensor tracks the wheel's rotational speed. The Tire-Road Interaction module assesses road conditions and the wheel's grip. The Machine Learning Control Algorithm section processes data from these sensors and, based on this information, governs the Brake Actuator through the Brake Actuator Valve. The implementation of this project involves utilizing MATLAB simulink toolbox in conjunction with Graphical User Interface (GUI) toolboxes to effectively realize the system.
\subsection{MATLAB SIMULINK}
\begin{figure}[H]
  \centering

```
    \includegraphics[height=15cm,width=08cm,keepaspectratio]{Picture9.png}
    \caption{Simulink Model design of PID and LQR based ABS}
    \label{}
\end{figure}
```

The MATLAB simulink design includes both a PID (Proportional – Integral – Derivative) controller-based Vehicle Antilock Braking System and an LQR (Linear Quadratic Regulator) based Vehicle Antilock Braking System, as shown in the figure 2.

```
\begin{figure}[H]
    \centering
    \includegraphics[height=15cm,width=08cm,keepaspectratio]{Picture11.png}
    \caption{PID Controller Vehicle Antilock Braking System Model}
    \label{fig:Binary format of Data sets}
\end{figure}
```

The figure 3 illustrates a Vehicle Antilock Braking System (ABS) featuring a PID (Proportional-Integral-Derivative) controller. Input variables such as Vehicle RPM, Tire diameter, a control block, and a timer are integral to this system's operation. An embedded system plays a pivotal role in calculating critical parameters like wheel speed, vehicle speed, and traveled distance. This embedded system operates within the realm of machine learning, where Embedded Machine Learning, often referred to as TinyML, comes into play. It empowers the deployment of machine learning models on resource-constrained nodes, offering unique opportunities for data-driven decision-making.

```
\begin{figure}[H]
    \centering
    \includegraphics[height=15cm,width=08cm,keepaspectratio]{Picture12.png}
    \caption{LQR (linear quadratic regulator) control method based Anti-braking system}
    \label{fig:Result Page}
\end{figure}
```

The figure 4 depicts a Vehicle Antilock Braking System (ABS) employing the Linear Quadratic Regulator (LQR) control, with MATLAB script coding utilized for LQR tuning and control. The system operates on a state-space model represented in the figure, receiving LQR data from the MATLAB script code. This system takes the actual speed as input, applies it to the LQR state-space model, and generates a speed-controlled output, which is then displayed on a scope.

```
\begin{figure}[H]
    \centering
    \includegraphics[height=15cm,width=08cm,keepaspectratio]{Picture13.png}
    \caption{Vehicle’s Anti-braking distance calculation model}
    \label{fig:: Result Page }
\end{figure}
```

The figure 5 provides a visual representation of the distance calculation process within a vehicle's anti-lock braking system, featuring both PID and LQR controllers. Within the realm of MATLAB, the Embedded coding segment assumes a central role in the definition and management of how model data is presented within the generated code. This greatly simplifies the software integration process.

```
\section{ RESULTS}
\subsubsection{Wheel Speed calculation:}
```

Wheel speed graph is updated on scope. This is using RPM wheel speed is calculated and updated

using embedded system tool box. Simulation time 0 to 500 taken to run and plot the graph.

```
\begin{figure}[!ht]
   \centering
   \includegraphics[scale=0.20]{wheelspeed.png}
   \caption{Wheel Speed Calculation}
\end{figure}
\subsubsection{PID Vehicle Speed and anti-braking system:}
```

PID controller-based vehicle speed controlling and anti-braking system is shown in the figure. Simulation time 0 to 500 taken to run and plot this speed tuning graph. Whenever speed increases and PID tuning speed of the vehicle linearly and Whenever speed decreases and PID tuning speed of the vehicle linearly and goes to stop stage smoothly.

```
\begin{figure}[!ht]
   \centering
   \includegraphics[scale=0.20]{1.png}
   \caption{PID Vehicle Speed and anti-braking system}
\end{figure}
\pagebreak
\subsubsection{PID with actual speed:}
```

This graph shows the comparison of actual speed of vehicle and PID controlled speed of the vehicle. Yellow line showing actual speed of the vehicle and blue line showing PID controlled speed of vehicle.

```
\begin{figure}[!ht]
   \centering
   \includegraphics[scale=0.20]{2 .png}
   \caption{PID with actual speed}
\end{figure}
\subsubsection{LQR Controller:}
```

LQR controller-based vehicle speed controlling and anti-braking system is shown in the figure. Simulation time 0 to 500 taken to run and plot this speed tuning graph. Whenever speed increases and LQR controller tuning speed of the vehicle linearly and whenever speed decreases and LQR controller tuning speed of the vehicle linearly and goes to stop stage smoothly.

```
\begin{figure}[!ht]
   \centering
   \includegraphics[scale=0.20]{3.png}
   \caption{LQR Controller}
\end{figure}
\subsubsection{Comparison of PID and LQR:}
```

This graph shows the comparison of LQR controlled speed of vehicle and PID controlled speed of the vehicle. Yellow line showing PID controlled speed of the vehicle and blue line showing LQR controlled speed of vehicle.

```
\begin{figure}[!ht]
   \centering
   \includegraphics[scale=0.50]{4.png}
   \caption{Comparison of PID and LQR}
\end{figure}
\pagebreak
```

```
\subsubsection{Speed Comparison of PID, LQR and Actual:}

This graph shows the comparison of actual speed, LQR controlled speed of vehicle and PID
controlled speed of the vehicle. Yellow line graph showing actual speed of the vehicle, blue line
graph showing PID controlled speed of vehicle and red line graph showing LQR controller speed
of the vehicle. Simulation conduction time is 0-500.
\begin{figure}[ht]
   \centering
   \includegraphics[scale=0.20]{5.png}
   \caption{Speed of PID,LQR and Actual}
\end{figure}
\subsubsection{Vehicle travelled distance:}
Vehicle travelled distance is updated in the below graph. Vehicle start time is 0 and vehicle stop
time 400 which mention and controlled by embedded system.
\begin{figure}[!ht]
   \centering
   \includegraphics[scale=0.20]{vehicletravelleddistance.png}
   \caption{Vehicle travelled distance}
\end{figure}
\pagebreak
\subsubsection{Main Window MATALB GUI App:}

The frontend system of the vehicle anti-lock braking system is depicted in the figure. It features
user-friendly controls, including a smart button for vehicle start and stop selection within the
GUI app. The "Simulate" button initiates the simulation process, providing insightful graphs
showcasing wheel speed, total vehicle distance traveled, and a comparative analysis of the anti-
lock braking system's performance.
\begin{figure}[!ht]
   \centering
   \includegraphics[scale=0.40]{mainwindow.png}
   \caption{Main Window MATALB GUI App}
\end{figure}

\subsection{Comparison Table}

\begin{figure}[H]
   \centering
\includegraphics[height=20cm,width=07cm,keepaspectratio]{pidlqr.png}
   \caption{Comparison Table }
   \label{fig:Comparison Table}
\end{figure}
 Final comparison results between PID Controller and LQR controller is tabulated above.


\section{Conclusion}
The modeling and control of the Anti-lock Braking System (ABS) have been successfully
executed. This simulation was accomplished using MATLAB simulink and various Graphic User
Interface (GUI) toolboxes. The ABS plays a pivotal role in enhancing the safety of modern
vehicles, ensuring their stability and reliability even at high speeds.
  \newline
The utilization of machine learning-based ABS offers improved efficiency compared to
```

```
conventional ABS systems. Two distinct control methods, namely the PID controller and LQR
controller, were employed for the system's control. These methods were rigorously compared,
with the PID controller and Embedded system emerging as the most effective control approach
for the vehicle's anti-lock braking system.
\subsection{Future Scope}
In the context of human-machine interaction, the design of intuitive and user-friendly interfaces
for ABS systems is paramount. Future research can delve into the development of human-
machine interface (HMI) solutions that provide real-time feedback to drivers, helping them better
understand and trust the ABS operation. Such interfaces can contribute to safer and more
effective utilization of ABS technology.
  \newline
Additionally, ABS research should consider environmental considerations. Researchers can
explore materials and manufacturing processes that minimize the environmental impact of ABS
components, aligning with broader sustainability goals in the automotive sector. This includes
assessing the life cycle of ABS systems to identify areas for improvement in terms of emissions
and resource usage.

    % \newpage
\addcontentsline{toc}{chapter}{References}
\printbibliography[title={References}]


\end{document}
```

CODE: <u>REPORT</u>

```
\documentclass[a4paper,12pt]{report}
\usepackage[utf8]{inputenc}
\usepackage{graphicx}
\usepackage{amsmath}
\usepackage{multirow}
\setcounter{secnumdepth}{3}
\usepackage{pdfpages}
\usepackage[nottoc,notlof,notlot]{tocbibind}
\renewcommand\bibname{References}
\usepackage{cite}
\usepackage{color}
\usepackage{algorithm,algpseudocode}
\renewcommand{\baselinestretch}{1.3}
\usepackage{enumitem}


\begin{document}
\tableofcontents
\listoffigures
\listoftables
\pagebreak
\pagenumbering{arabic}
\chapter{Early Cataract Detection with Deep Learning-Based Computer-Aided Systems}
```

```
\section{Introduction}
Cataracts are a common eye disease that affects millions of people worldwide. A cataract occurs when the natural lens of the eye becomes clouded, leading to blurry vision and, if left untreated, blindness. Early detection and treatment of cataracts are crucial to prevent blindness.
Currently, the detection of cataracts is typically done through a comprehensive eye exam conducted by a trained ophthalmologist. However, this process is time-consuming and requires specialized training, making it difficult to provide timely and accurate diagnoses, particularly in areas with limited access to healthcare.
In recent years, deep learning has emerged as a powerful tool in medical imaging analysis. Deep learning techniques, such as MLP, CNN, and VGG16, can automatically learn and extract relevant features from medical images, allowing for faster and more accurate diagnoses.

In this report, we present CataractNet, an automated cataract detection system using deep learning for fundus images. We used a combination of MLP, CNN, and VGG16 for feature extraction and classification. We also selected SVM and Random Forest as the classification algorithms. The system was trained and evaluated on a dataset of fundus images.
The goal of this report is to evaluate the performance of CataractNet in detecting cataracts in fundus images and to demonstrate its potential for assisting ophthalmologists in diagnosing cataracts at an early stage, particularly in areas with limited access to healthcare. We believe that CataractNet can improve the quality of healthcare and reduce the burden on ophthalmologists by providing fast and accurate diagnoses of cataracts.
%Write a brief introduction of your project, add  citation \cite{journal1}.\left(

\section{Motivation}
The primary motivation of this project is to develop an accurate and efficient system for cataract detection that utilizes the power of deep learning techniques. By combining MLP, CNN, and VGG16 for feature extraction and classification, we aim to achieve the detection of nuclear cataracts with higher accuracy and reliability in cataract diagnosis.
%write motivation of your project

\section{Problem Statement}
The problem this project aims to address is the accurate and timely diagnosis of nuclear cataracts using deep learning techniques. Cataracts are a common eye condition that affects people of all ages, but especially older adults, and early detection is crucial for effective management. However, cataract diagnosis and management can be challenging for ophthalmologists due to the variability of the disease's presentation.

\section{Objectives}
\begin{itemize}
\item To develop an automated system for cataract detection using deep learning techniques that can accurately detect the presence and severity of cataracts and nuclear cataract detection in patients' eyes.

\item To improve the accuracy and reliability of cataract diagnosis by combining MLP, CNN, and VGG16 for feature extraction and classification.

\item To evaluate the performance of the proposed system by comparing it to existing cataract detection methods, such as manual diagnosis by ophthalmologists.

\item To develop a user-friendly interface for the system that can be easily used by ophthalmologists and other healthcare professionals.
\end{itemize}
```

```
\section{Project Outcome and Mode of Demonstration}
The project outcome will be an automated system for cataract detection that utilizes deep learning techniques such as MLP, CNN, and VGG16 for feature extraction and classification. The system will take as input an image of a patient's eye and output a diagnosis of the presence and severity of cataracts. The system will be designed to be accurate, reliable, and user-friendly, allowing ophthalmologists and other healthcare professionals to make informed decisions about patient care.

The mode of demonstration of the project outcome could be through a software application that demonstrates the system's functionality and performance. The demonstration could include the system's user interface, which should be intuitive and easy to use, as well as examples of the system's output for different types and severities of cataracts.

\section{Applications}
\begin{itemize}
\item Clinical diagnosis: The system can be used by ophthalmologists and other healthcare professionals to assist in the diagnosis of cataracts. It can provide a reliable and objective assessment of the presence and severity of cataracts, leading to earlier detection and timely treatment.
\item Screening programs: The system can be used as part of screening programs to identify individuals at risk of developing cataracts. It can help to detect cataracts at an early stage, allowing for early intervention and preventing vision loss.
\item Telemedicine: The system can be used in telemedicine applications, where ophthalmologists can remotely assess patients' eye images for cataract diagnosis. This can be especially beneficial in areas where access to specialized healthcare services is limited.
\item Research: The system can be used as a tool for research purposes, such as studying the prevalence and risk factors of cataracts in different populations or evaluating the effectiveness of different treatment options.
\end{itemize}
\pagebreak

\chapter{Literature Survey}
The study demonstrates the potential of deep learning and machine learning algorithms for automated cataract detection. The proposed system, CataractNet, builds upon these studies by using a combination of MLP, CNN, and VGG16 for feature extraction and classification, and SVM and Random Forest algorithms for classification.
\section{Previous Research}
In recent years, researchers have explored various methods for automated cataract detection using deep learning and machine learning algorithms. Here are some key studies in this field:
\begin{itemize}
  \item "Cataract detection using deep learning: A survey" by Tan and Acharya (2020) - This survey paper provides an overview of recent developments in automated cataract detection using deep learning. The authors review various techniques for feature extraction, classification, and segmentation of cataractous regions in fundus images.
  \item "Cataract detection using morphology-based segmentation and features from spatial gray-level dependence matrices and color histograms" by Zhang et al. (2017) - In this study, the authors propose a system for cataract detection using morphology-based segmentation and features from spatial gray-level dependence matrices and color histograms. The system achieves an accuracy of 86.7\% on a dataset of 450 fundus images.
  \item "Cataract detection using wavelet-based features and support vector machine classification" by Zhang et al. (2018) - In this study, the authors propose a system for cataract
```

detection using wavelet-based features and support vector machine (SVM) classification. The system achieves an accuracy of 95.7\% on a dataset of 450 fundus images.

\item "Automatic detection of cataract using convolutional neural networks" by Kumar et al. (2017) - In this study, the authors propose a system for automatic detection of cataract using convolutional neural networks (CNN). The system achieves an accuracy of 94.5\% on a dataset of 1,000 fundus images.

\item "Cataract diagnosis using a random forest classifier with dynamic features" by Jiang et al. (2019) - In this study, the authors propose a system for cataract diagnosis using a random forest classifier with dynamic features. The system achieves an accuracy of 88.3\% on a dataset of 1,200 fundus images.

\item "Automatic detection and grading of cataract using a combination of deep learning and handcrafted features" by Yan et al. (2020) - In this study, the authors propose a system for automatic detection and grading of cataract using a combination of deep learning and handcrafted features. The system achieves an accuracy of 94.7\% on a dataset of 1,200 fundus images.

\end{itemize}

\section{Summary of Literature Review}

In recent years, there has been significant research in the development of automated systems for cataract detection using deep learning techniques.Several studies have proposed novel approaches for cataract detection using various deep learning architectures such as CNNs, MLPs, and ensemble learning approaches. The use of pre-trained models such as VGG16 and transfer learning has also been explored in cataract detection.As seen in table \ref{my-label}.

\begin{table}[htbp]
\centering
\caption{Literature Survey}
\label{tab:litsurvey}
\resizebox{\columnwidth}{!}{%
\begin{tabular}{|p{7cm}|p{1cm}|p{5cm}|p{7cm}|}
\hline
\textbf{Title} & \textbf{Year} & \textbf{Methodology} & \textbf{Objectives} \\
\hline
Automated detection of cataracts in adult human eyes using deep learning & 2018 & Pre-trained CNN & To develop an automated system for cataract detection using deep learning and achieve high accuracy. \\
\hline
Cataract detection using deep convolutional neural network with transfer learning & 2019 & Pre-trained CNN with transfer learning & To propose a deep learning system for cataract detection using transfer learning and achieve high accuracy. \\
\hline
A novel approach for cataract detection and classification using deep learning techniques & 2020 & MLP and CNN combination & To propose a novel approach for cataract detection and classification using a combination of MLP and CNN architectures. \\
\hline
Automated diagnosis of cataract using ensemble learning approach & 2021 & Ensemble learning & To propose an automated diagnosis system for cataract using an ensemble learning approach and achieve high accuracy. \\
\hline
\hline
Automatic diagnosis of cataracts based on deep learning and feature fusion of slit-lamp images & 2018 & Pre-trained VGG16 & To propose an automated system for cataract diagnosis based on deep learning and feature fusion of slit-lamp images. \\
\hline

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\hline
Cataract detection using machine learning algorithms: A comparative study & 2020 & Machine learning algorithms & To compare the performance of different machine learning algorithms for cataract detection. \\
\hline
\hline
Deep learning for automatic cataract detection and classification from retinal images & 2021 & CNN & To develop a deep learning-based system for cataract detection and classification from retinal images. \\
\hline
\end{tabular}
}\end{table}
Overall, the studies demonstrate the potential of deep learning techniques for cataract detection, achieving high levels of accuracy and reliability. Automated cataract detection can improve patient outcomes and quality of life by enabling early detection and treatment. The literature review also highlights the need for further research in the field, particularly in developing automated systems that can handle larger datasets and can accurately diagnose different types and severities of cataracts.
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\chapter{Requirements}
The proposed system, CataractNet, is an automated cataract detection system using deep learning for fundus images. The system uses a combination of MLP, CNN, and VGG16 for feature extraction and classification.
\begin{enumerate}[label=\roman*]
\item Preprocessing: The input fundus images are preprocessed to remove any noise or artifacts that may affect the performance of the system. The preprocessing steps include contrast enhancement, normalization, and resizing of the images to a standardized size.
\item Feature Extraction: The preprocessed images are fed into the MLP, CNN, and VGG16 models to extract features from the images. The MLP model extracts global features, while the CNN and VGG16 models extract local features from different levels of the images.
\item Classification: The extracted features are then fed into SVM and Random Forest algorithms for classification. The algorithms are trained on a labeled dataset of fundus images to distinguish between normal and cataractous images.
Performance Evaluation: The performance of the proposed system is evaluated by comparing its results with the ground truth labels. The evaluation metrics used include accuracy, sensitivity, specificity, and F1-score.
\end{enumerate}
\section{Proposed Block Diagram}
\begin{figure}[h]
        \center
        \includegraphics[width=1.0\textwidth]{block diagram.jpg}
        \caption{Block Diagram}
        \label{fig1}
\end{figure}

you can call the figure number  whenever you required ex. As seen in Figure \ref{fig1}

\section{Software Requirements}
\begin{itemize}
\item IDE : FLASK
```

```
\item Language : Python
\item Tool : Jupyter Notebook
\item Software : Anaconda
\item Front End : HTML, CSS
\item Libraries : Tensorflow, keras, numpy,pandas
\end{itemize}
\section{Hardware Requirements}
\begin{itemize}
   \item RAM : 2 GB
   \item Hard disk       : 100 GB
   \item Process : 32/64 Pentium
\end{itemize}
\section{Methodology}
\begin{itemize}
   \item Data Preprocessing: The input data, consisting of images of patients' eyes, will be preprocessed to remove noise, normalize the illumination, and resize the images to a uniform size.
   \item Feature Extraction: The preprocessed images will be passed through a combination of deep learning architectures such as MLP, CNN, and VGG16 to extract features that are relevant for cataract detection.
   \item Classification: The extracted features will be fed into a classification algorithm, such as a support vector machine (SVM) or a random forest classifier, to classify the images into cataract or non-cataract classes.
   \item Evaluation: The performance of the system will be evaluated by comparing the output of the system with ground-truth labels using metrics such as accuracy, precision, recall, and F1 score.
   \item Deployment: The final step is to deploy the system for real-world use. This could involve integrating the system with a web application or mobile app that can be used by healthcare professionals or patients to detect cataracts.
\end{itemize}
\section{Novelty of proposed work}
In this project, machine learning and deep learning algorithms, specifically CNN (Convolutional Neural Network), VGG16 (a popular pre-trained CNN model), and MLP (Multilayer Perceptron), will be utilized for prediction purposes. These algorithms offer novel and powerful capabilities for analyzing and understanding complex patterns and relationships within data. By combining these machine learning and deep learning algorithms—CNN, VGG16, and MLP—this project can leverage their unique strengths and capabilities to tackle diverse prediction tasks, offering a comprehensive and robust approach to predictive modeling across various domains.

% to place figure at top the page
%\begin{figure}[!t]
%       \center
%       \includegraphics[width=0.5\textwidth]{images/Fig1.png}
%       \caption{image caption}

%\end{figure}

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\chapter{Planning and Feasibility Of Work}
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write Some things here.....
\section{Feasibility Analysis}
\textbf {Technical Feasibility:}
\begin{itemize}
\item  Availability of a dataset: A large and diverse dataset of cataract and non-cataract images is required to train and test the system.
\item  Adequate computing resources: Developing and training deep learning models requires significant computing power and memory.
\item  Expertise in deep learning: Developing a deep learning system requires expertise in programming, mathematics, and machine learning.
\end{itemize}
\textbf{Economic Feasibility:}
\begin{itemize}
   \item Budget: The development of a cataract detection system using deep learning techniques may require a significant investment in terms of hardware, software, and personnel.
   \item Return on Investment: The potential benefits of an automated cataract detection system, such as improved patient outcomes, reduced healthcare costs, and increased efficiency, may outweigh the costs.
\end{itemize}
\textbf{Legal and Ethical Feasibility:}
\begin{itemize}
   \item Data Privacy: It is essential to ensure that patient data is anonymized and protected to avoid breaches of privacy and confidentiality.
   \item Regulatory compliance: Developing a medical application requires compliance with regulatory standards, such as the US Food and Drug Administration (FDA) and the European Medicines Agency (EMA).
   \item Ethical considerations: Ensuring that the system does not perpetuate biases or discriminate against vulnerable populations.
\end{itemize}
\section{Anticipated Bottleneck        }
\begin{itemize}
   \item Insufficient Data: The availability of a large and diverse dataset is crucial for the development of a robust cataract detection system. Insufficient or low-quality data can limit the accuracy and reliability of the system.
   \item Overfitting: Overfitting occurs when the system is trained on a limited dataset and performs poorly on new, unseen data. It can be mitigated by techniques such as regularization and data augmentation.
   \item Hardware Limitations: Developing and training deep learning models requires significant computing power and memory. Limited hardware resources can slow down the development process and limit the complexity of the models.
   \item Algorithm Complexity: Developing complex algorithms using deep learning techniques can be time-consuming and computationally expensive. It may be necessary to simplify the models to achieve acceptable performance levels.
\end{itemize}
\section{Budget Estimation}
Budget of your project work
\section{Pert Chart}

\begin{figure}[h]
        \center
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        \includegraphics[width=1.0\textwidth]{PERT CHART.png}
        \caption{Pert Chart}
        \label{fig2}
\end{figure}

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